

COMS5026A - Applied Machine Learning

Lab Exercise 2

The second lab assignment is to use `scikit-learn` to implement a neural network for the Iris dataset, which is available on the Moodle page.

1. Construct a neural network consisting of:

- an input layer with 4 nodes,
- at least one hidden layer (with, say, 20 nodes),
- an output layer with 3 nodes.

You may add more hidden layers if you wish, but at least one is required.

2. Prepare the Iris dataset by:

- (a) normalising the input values using max-min normalisation;
- (b) creating a one-hot encoding of the target values.

3. Split the dataset into training, validation, and test sets.

4. Train the neural network on the prepared Iris dataset. Use the **sum-of-squares** loss function.

5. During training, after every epoch, print:

- the sum-of-squares loss on the training set,
- the sum-of-squares loss on the validation set.

6. After training, print the accuracy on the test set.

Required: You must demonstrate to a tutor that your code works and prints out the required values. In addition, you must show that you know how to change the neural network structure. You may demonstrate to any of the tutors at the lab. This must be done during next week's lab (**before 12h00 on June 4**).

Lab assignments count towards the final mark for the course.